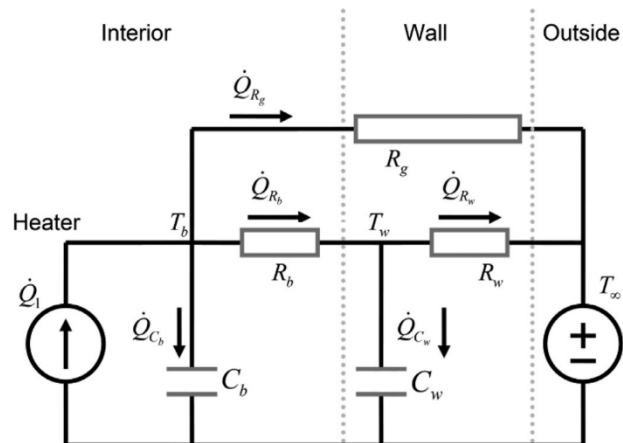


Interview Task Description: Building Thermal Modelling

Background:

Heating and cooling of buildings for thermal comfort constitutes a major part of the world-wide energy demand. Energy Management Systems (EMS) seek to automatically reduce the energy consumption of buildings or use their thermal inertia to shift electric load. A precondition for EMS control is a sufficiently accurate model of building thermal dynamics, i.e., how the indoor temperature evolves based on weather conditions and heating/ cooling power applied. There are different ways to model thermal dynamics based on different information and/ or data available. If only measurement data and no information about the physical characteristics of the building is available, data-driven methods can be applied to model thermal dynamics. One type of data-driven model is Resistor-Capacitor equivalent circuit models (short “R-C models”). An example of an R-C model taken from [1] is shown at the top right. We have provided you with a file containing 2 years’ worth of measurements from a single-family residential building located in London that can be used for data-driven modelling.



[1] Brastein, O. M.; Perera, D.W.U.; Pfeifer, C.; Skeie, N.-O. (2018): Parameter estimation for grey-box models of building thermal behaviour. In *Energy and Buildings* 169, pp. 58–68. DOI: 10.1016/j.enbuild.2018.03.057.

Task:

The following questions shall be answered:

1. Regarding the data contained in the file, which characteristics and differences does the building data reveal when comparing the first and second year? What could be the causes? How can this be visualized?
2. How can R-C models of different complexity predicting the indoor temperature based on outside weather and heating power be developed? Please train several R-C models with different complexity using data from year 1 and then use data from year 2 for testing the models.
3. Calculate the prediction performance of your models for the first 10 days of year 2 (winter) and the 10 days starting on the 200th day of year 2 (summer), considering different inference horizons. Which model performs best and why?

Please prepare a presentation of about 15 minutes with slides on the tasks above. Your presentation should include an explanation of your approach and a short discussion of your results.